

Hoang Nguyen

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Cambridge, MA

Education

Massachusetts Institute of Technology (MIT) , Cambridge, MA, USA <i>Ph.D., Electrical Engineering and Computer Science</i>	2024–Present
Chung-Ang University (CAU) , Seoul, South Korea <i>M.E., Electrical and Electronics Engineering</i> Graduated with Dean’s Award	2021–2023
Hanoi University of Science and Technology (HUST) , Hanoi, Vietnam <i>B.E., Control Engineering and Automation</i> Valedictorian – Rank: 1/297	2015–2020

Research Experience

Laboratory for Information and Decision Systems , MIT <i>Research Assistant, advised by Prof. Priya Donti</i>	2024–Present
Machine Decision Intelligence and Learning Lab , KAIST <i>Research Scientist, advised by Prof. Donghwan Lee</i>	2023–2024
Electric Power System and Smart Grid Lab , CAU <i>Research Assistant, advised by Prof. Dae-Hyun Choi</i>	2021–2023
Department of Automatic Control , HUST <i>Research Assistant, advised by Prof. Nga Thi-Thuy Vu & Prof. Phuong Nam Dao</i>	2017–2020

Publications

- [1] **H. T. Nguyen** and P. Donti, “FSNet: Feasibility-seeking neural network for constrained optimization with guarantees,” *Advances in neural information processing systems*, vol. 38, pp. 39 670–39 708, 2026, (featured on [MIT News](#)).
- [2] **H. T. Nguyen**, S. Liu, R. S. Biswas, V. Pendyala, N. Virani, D. Deka, and P. L. Donti, “Scalable self-supervised learning for multiphase AC-OPF in distribution systems with topology reconfiguration,” *arXiv preprint arXiv:2608.25095*, 2026.
- [3] **H. T. Nguyen** and D.-H. Choi, “Distributionally Robust Model Predictive Control for Smart Electric Vehicle Charging Station with V2G/V2V Capability,” *IEEE Transactions on Smart Grid*, vol. 14, no. 6, pp. 4621–4633, 2023.
- [4] **H. T. Nguyen** and D.-H. Choi, “Distributionally Robust Safety Filter for Learning-Based Control in Active Distribution Systems,” *IEEE Transactions on Smart Grid*, vol. 14, no. 6, pp. 4972–4975, 2023.
- [5] **H. T. Nguyen**, Y.-J. Kim, and D.-H. Choi, “Sample-Efficient Learning for a Surrogate Model of Three-Phase Distribution System,” *IEEE Transactions on Power Systems*, vol. 39, no. 1, pp. 2361–2364, 2023.
- [6] S. Jeon, **H. T. Nguyen**, and D.-H. Choi, “Safety-Integrated Online Deep Reinforcement Learning for Mobile Energy Storage System Scheduling and Volt/VAR Control in Power Distribution Networks,” *IEEE Access*, vol. 11, pp. 34 440–34 455, 2023.

- [7] **H. T. Nguyen** and D.-H. Choi, “Decentralized Distributionally Robust Coordination Between Distribution System and Charging Station Operators in Unbalanced Distribution Systems,” *IEEE Transactions on Smart Grid*, vol. 14, no. 3, pp. 2164–2177, 2022.
- [8] **H. T. Nguyen** and D.-H. Choi, “Three-Stage Inverter-Based Peak Shaving and Volt-VAR Control in Active Distribution Networks Using Online Safe Deep Reinforcement Learning,” *IEEE Transactions on Smart Grid*, vol. 13, no. 4, pp. 3266–3277, 2022.
- [9] **H. T. Nguyen**, H. Q. Nguyen, P. N. Dao, and N. T.-T. Vu, “On Robust Control of Permanent Magnet Synchronous Generators Using Robust Integral of Error Sign,” in *Computational Intelligence Methods for Green Technology and Sustainable Development: Proceedings of the International Conference GTSD2020 5*, Springer, 2021, pp. 156–166.
- [10] A. D. Hoang, H. Q. Nguyen, P. N. Dao, and **H. T. Nguyen**, “Output Feedback Adaptive Reinforcement Learning Tracking Control for Wheel Inverted Pendulum Systems,” in *Research in Intelligent and Computing in Engineering: Select Proceedings of RICE 2020*, Springer, 2021, pp. 535–547.
- [11] N. T. Dang, **H. T. Nguyen**, and P. N. Dao, “Modified Q-Learning Algorithm with Lifting Method for Discrete-Time Linear Periodic Systems,” in *International Conference on Engineering Research and Applications*, Springer International Publishing Cham, 2020, pp. 548–559.

Posters and Presentations

5th Workshop on Foundation Models of the Electric Grid <i>Self-supervised Learning for AC Optimal Power Flow in Distribution Systems with Hard Constraints</i>	Mar. 2026
MIT-GE Vernova Energy and Climate Symposium <i>AI for Large-Scale, Topology-Aware AC-OPF in Distribution Grids</i>	Feb. 2026
39th Annual Conference on Neural Information Processing Systems <i>FSNet: Feasibility-Seeking Neural Network for Constrained Optimization with Guarantees</i>	Dec. 2025

Awards & Honors

Jin Au Kong Fellowship, MIT	2024
CAU Graduate School Dean’s Award (for one outstanding Master’s student)	2023
CAU Young Scientist Scholarship	2021–2023
Valedictorian Award, HUST	2020
Undergraduate Scholarships and Awards	2015–2020
Vallet Scholarship (France), IHI Scholarship (Japan), Nitori Scholarship (Japan), Pham Van Tra Scholarship, Academic Excellence Scholarships, Best Student Awards	

Academic Service

Reviewer for: IEEE Transactions on Smart Grid, IEEE Transactions on Power Systems, 2023–Present
 IEEE Transactions on Sustainable Energy